FEM for 1D problems

Summary and discussion of results

Discretization error

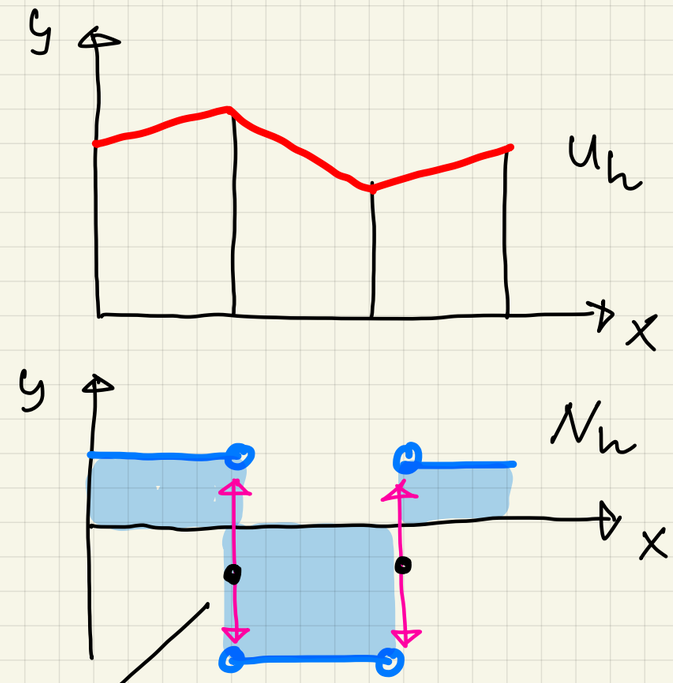
Error function

$$e = u - u_h$$

How to make sure the discretization error is sufficiently small

- Exact solution, but only available in simple cases
- Theory of error estimators (mathematicians)
- Rules of thumb and engineering expertise
  - Inspect stresses and jumps
  - Study results for different meshes

Axial force



u_h is piecewise linear

N_h is piecewise constant

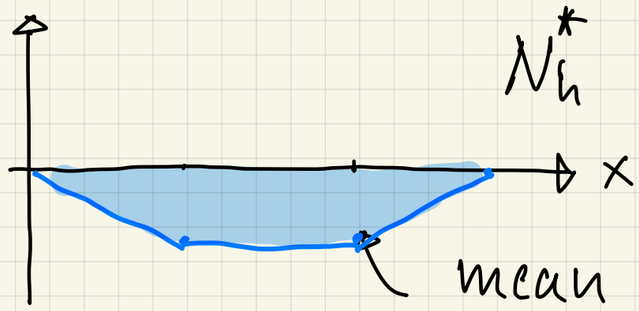
$$N(x) = EA\,u'(x)$$

In FE-programs: Element solution } Use that to assess the quality of your results

→ Jumps physically not possible

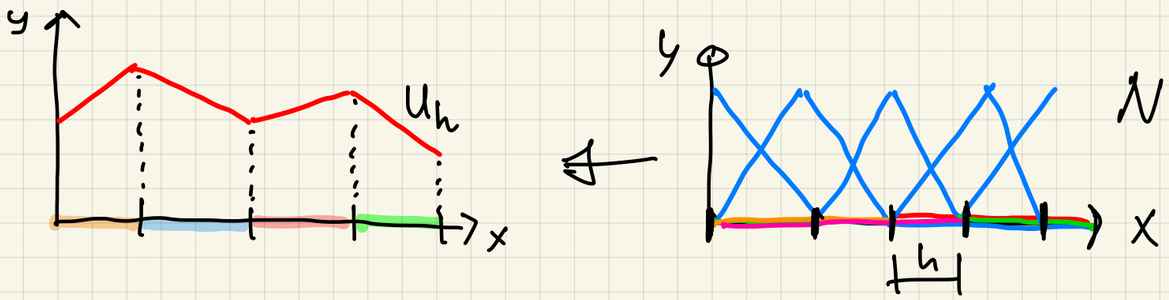
"Improved" plot

Nodal solution in FE-software } Show this to your clients

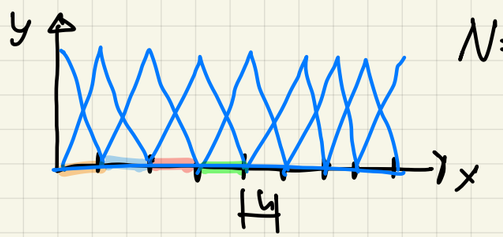


mean value of both elements

How to improve the approximate solution u_h

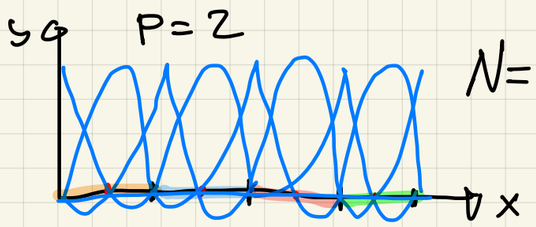


Reduce element size



h-version of finite element method

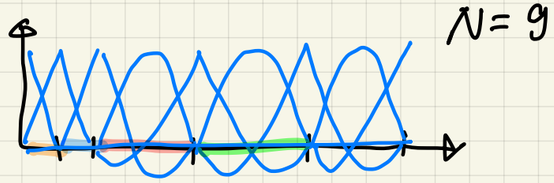
Increase polynomial degree



p-version of FEM

up to $p \cong 15$

Combination

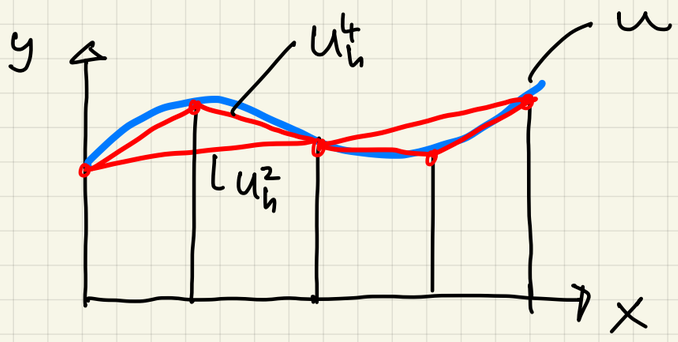


hp-version of FEM

Converges faster to exact solution than h-version but difficult to program

→ Better solution with same N

Convergence of solution



Sequence of functions u_h^k with

$$\lim_{k\to\infty}(u_h^k - u) = 0$$

Proof is difficult (we leave that to mathemachicians)

In our computation, we used the supremum-norm

$$\|f\|_\infty = \sup\{f(x) \mid 0 \le x \le L\}$$ (gleichmäßige Konvergenz)

More common: Energy-norm

$$\|f\|_a = \frac{1}{2}a(f,f) \qquad a(u,\delta u) = EA\int_0^l u'\delta u'\,dx + C\int_0^l u\,\delta u\,dx + S\cdot u(l)\cdot\delta u(l)$$

→ Computes energy associated with the error

Convergence plots

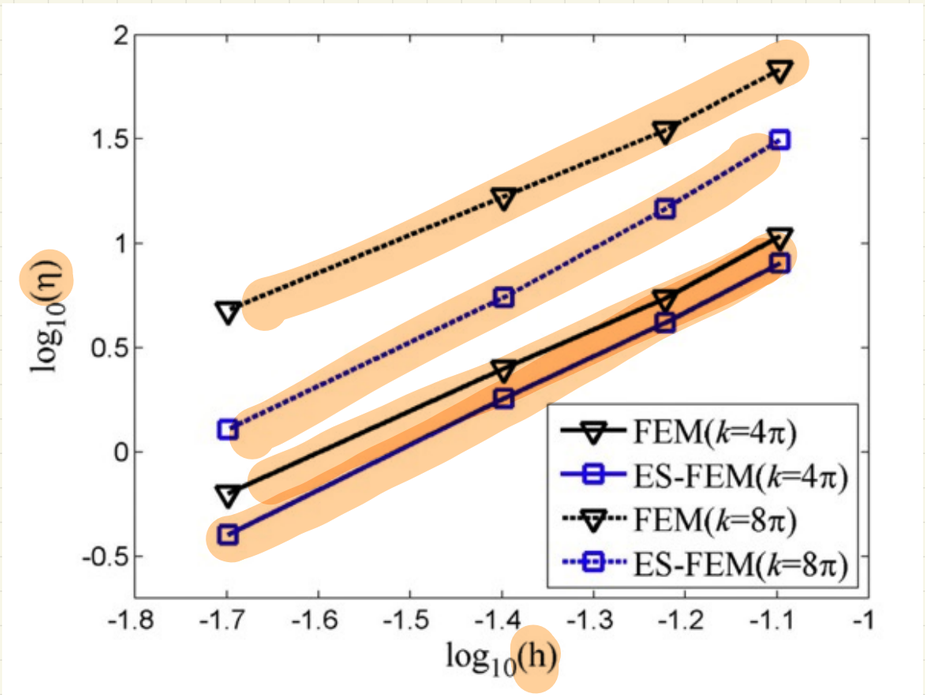
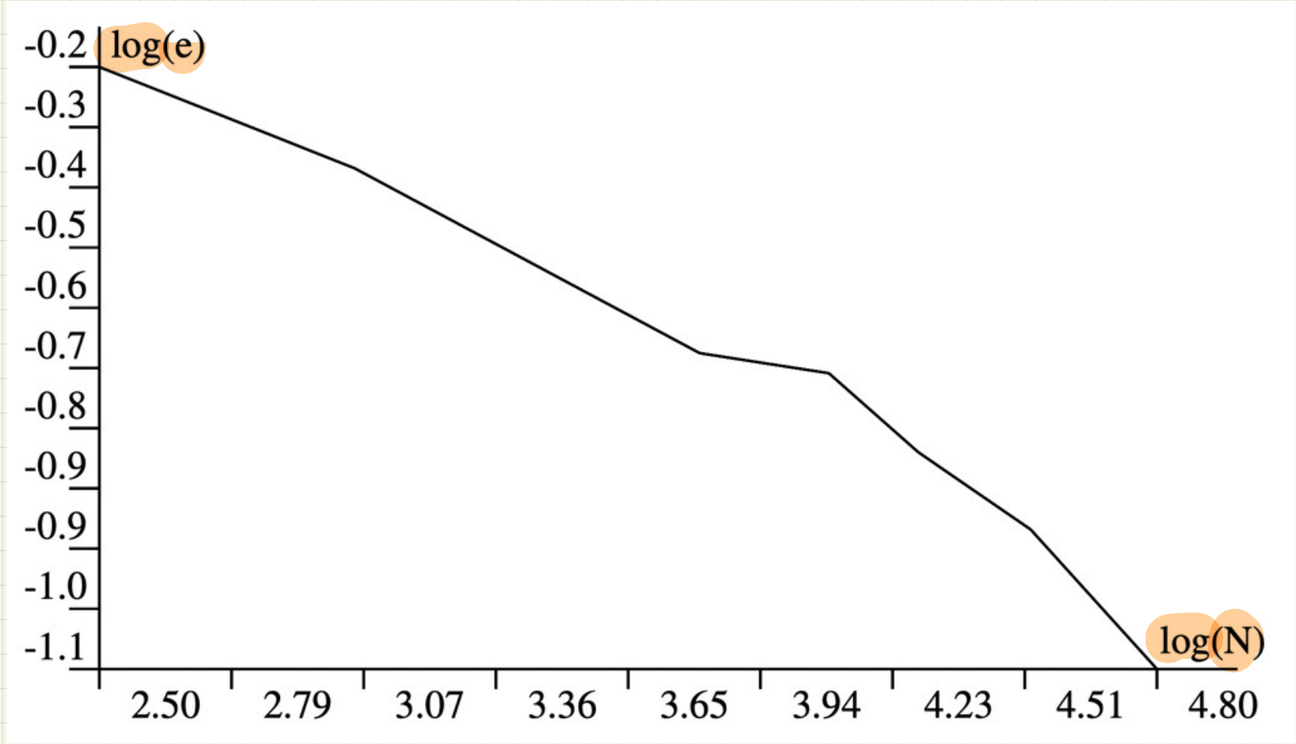


Fig. 9. The convergence rate results for FEM and ES-FEM.

Chai et al. (2018)



Rachowicz et al. (2006)